

ChatGPT

Congratulations! Once you have PDF ingestion, SQL Server, Qdrant, Ollama, and FastAPI all working together, you've built the foundation of a Retrieval-Augmented Generation (RAG) system.

Here are the features I'd recommend adding next, roughly in order of impact.

Phase 1: Better Document Management (High Priority)

1. File Upload API

Instead of sending a file path:

```
JSON
{
  "file_path": "app/data/documents/handbook.pdf"
}
```

allow users to upload directly:

```
POST /upload

Form-Data
file = handbook.pdf
```

Benefits:

- No local path required
- Works from a web UI
- Easier for end users

2. Document Dashboard

Create endpoints like:

```
GET /documents
```

Returns:

```
JSON
[
  {
    "id": 1,
    "filename": "handbook.pdf",
    "pages": 15,
    "chunks": 42,
    "uploaded": "2026-06-26"
  }
]
```

3. Delete Document

```
DELETE /documents/{id}
```

Deletes:

- SQL document
- SQL chunks
- Qdrant vectors

4. Re-index Document

```
POST /documents/{id}/reindex
```

Useful after changing chunk size or embedding model.

Phase 2: Better Search

5. Return Sources

Instead of:

```
JSON
{
  "answer": "Employees receive 20 days leave."
}
```

Return:

```
JSON
{
  "answer": "Employees receive 20 days leave.",
  "sources": [
    {
      "document": "handbook.pdf",
      "page": 5,
      "score": 0.94
    }
  ]
}
```

6. Page Numbers

Store page metadata:

```
Python
payload={
  "page": 5,
  "document": "handbook.pdf"
}
```

Now answers can cite:

Source:

handbook.pdf

Page 5

7. Similarity Score

Display confidence:

```
JSON
"score":0.96
```

Useful for deciding whether to answer or say:

I couldn't find information.

8. Hybrid Search

Instead of vector search only:

```
Vector Search
+
Keyword Search
```

Much more accurate.

Phase 3: Multiple File Types

Support:

- PDF 
- DOCX
- TXT
- Markdown
- CSV
- Excel
- PowerPoint
- HTML

Use document loaders based on extension.

Phase 4: Chat Memory

Current:

Printed using [ChatGPT to PDF](#), powered by PDFCrowd [HTML to PDF API](#).

Q:
What is leave policy?

A:
20 days.

Next:

Q:
Who approves it?

A:
Managers approve annual leave.

The system remembers context.

Store:

Conversation

↓

Messages

↓

History

Phase 5: User Authentication

Login

↓

JWT

↓

Roles

Admin

HR

Employee

Viewer

Permissions:

Admin

Upload

Delete

Re-index

Users

Only Chat

Phase 6: Admin Portal

Dashboard
Documents
Chunks
Embedding Count
Qdrant Health
Ollama Status
SQL Status
Recent Uploads
Search Logs

Phase 7: Modern Chat UI

Instead of Postman:

```
+-----+
| Knowledge Assistant |
+-----+

User:
How many leave days?

Assistant:
Employees receive 20 days annual leave.

Source:
handbook.pdf
Page 7
```

Like ChatGPT.

Phase 8: Streaming Responses

Instead of waiting:

```
Thinking....
(wait 5 sec)
```

Show:

```
Employees receive...
20 days...
annual leave...
```

Streaming feels much better.

Phase 9: OCR

Support scanned PDFs.

Example:

```
Image PDF
↓
OCR
↓
Text
↓
Embeddings
```

Phase 10: AI Improvements

Instead of:

```
Answer ONLY from context.
```

Use prompts like:

```
You are an HR Assistant.

Only answer using retrieved documents.

If information is unavailable,
say "The knowledge base does not contain that information."

Always cite the document name and page.
```

Phase 11: Analytics

Track:

```
Most asked questions
Popular documents
Search failures
Average response time
Token usage
Embedding count
```

Great for improving the knowledge base.

Phase 12: SQL Improvements

Instead of only:

Documents
Chunks

Store:

Documents
Chunks
Conversations
Messages
Users
Roles
SearchLogs
Feedback
Embeddings
AuditLogs

Phase 13: Feedback

After each answer:

👍 Helpful

👎 Not Helpful

Store:

Question
Answer
Feedback
Time
User

Phase 14: Multiple Collections

HR
Finance
IT
Engineering
Sales

Each with its own Qdrant collection.

Phase 15: Multi-Agent

Instead of one assistant:

HR Agent
IT Agent
Finance Agent
Legal Agent
Policy Agent

A router decides which agent should answer.

Suggested Roadmap

Here's the order I'd recommend:

1. ☒ File upload API
2. ☒ Chat web interface (React or HTML/JavaScript)
3. ☒ Source citations with page numbers
4. ☒ Conversation history
5. ☒ Authentication and user management
6. ☒ Support for DOCX, Excel, and other file formats
7. ☒ Admin dashboard with document management
8. ☒ Streaming responses
9. ☒ Hybrid search (vector + keyword)
10. ☒ Feedback and analytics

Following this roadmap will take your current prototype to a production-ready enterprise knowledge assistant suitable for HR, IT support, policies, SOPs, and internal documentation.